Solstice Task 2 - PART C

Mathematics Club

Instructions

- Provide descriptive answers for all questions. Explain your thought process well.
- Having solved this task and having the feel of it will be very satisfying, full trust. Enjoy! (From prior experiences)
- It is suggested and expected to not use LLMs (ChatGPT, Gemini, etc.) for thinking purposes. Feel free to use it for understanding purpose but not blindly. If the question asks you to derive/prove something, have fun struggling solving the problem and feeling the joy as you get closer to the answer. If you find yourself utterly stuck, do ask for hints on group or DM!

All the best and have fun!

§1 Fisher Information Matrix

1. Show that when evaluated at a point on the set of true parameters $w^{(0)} \in W_0$, the FIM $I(w)$ is

$$I_{j,k}\left(w^{(0)}\right) = \frac{\partial^2}{\partial w_j \, \partial w_k} K(w)\bigg|_{w=w^{(0)}}$$

   Hints:

   a) Start with $K(w)$. Differentiate w.r.t. some w_j. Use $\dfrac{\partial}{\partial w_j} \log f(w) = \dfrac{1}{f(w)} \dfrac{\partial f(w)}{\partial w}$. Play with integrals by using what depends on w_j and what is independent.

   b) Take second derivative w.r.t. w_k. Use:

$$\frac{\partial^2 \log p}{\partial w_j \, \partial w_k} = \frac{1}{p} \frac{\partial^2 p}{\partial w_j \, \partial w_k} - \left(\frac{\partial \log p}{\partial w_j}\right)\left(\frac{\partial \log p}{\partial w_k}\right)$$

   c) While evaluating at $w^{(0)}$, one of the term cancels out and what will remain is $I_{j,k}(w^{(0)})$.

2. What is meant by a positive definite matrix? Prove that if a square matrix has elements from $\mathbb{R}$, is positive definite and symmetric, then it has positive eigen values.
   Hint: Linal, better call Hemanth to explain spectral decomposition.

3. What is meant by rank of a matrix? If a matrix is not full rank, prove that $\det I(w^0) = 0$.

4. Show that for a regression model, $I(w^{(0)})$ is degenerate iff the set of derivatives

$$\left\{\frac{\partial}{\partial w_j} f(x, w)\right\}_{j=1}^{d}$$

   is linearly dependent.

5. Given an example of a two layer neural network using a ReLU function and prove how it is singular.

§2 Indeed all regular things are normal

Asymptotic normality: The posterior of regular models converges in distribution to a d-dimensional normal distribution centred at the maximum likelihood estimator $w^{(0)}$:

$$p(w \mid D_n) \to \mathcal{N}_d\left(w^{(0)}, \frac{1}{n} I\left(w^{(0)}\right)^{-1}\right).$$

Prove the above. *Hints:*

1. Write the posterior equation. Take log.
2. Take the second order Taylor expansion around the MLE $\hat{w_n}$
3. What can u say about the first derivative?
4. What "Information" can you say about the second derivative?
5. What can I tell about the log of prior? Make some assumptions.
6. Took log previously, go back to exp, put things together. Normalize. Done!

§3 Bayesian Information Criterion

Bayesian Information Criterion (BIC): The free energy of regular models asymptotically looks like the BIC as $n \to \infty$, where $L_n\left(w^{(0)}\right) = \min_{w \in W} L_n(w)$ and d is the dimension of parameter space $W \subseteq \mathbb{R}^d$:

$$F_n \approx nL_n\left(w^{(0)}\right) + \frac{d}{2}\log n = \text{BIC}.$$

This is not very hard to prove. Try to do it without help of blog. Reflect on $\det I$.

Go to blog and think of what do they even want to say by calling it accuracy-complexity tradeoff? *Hints:*

1. Fisher information needs to be used. It looks like it looks something like a second derative. So we Taylor expand something – take the Negative log likelihood around true parameter $w^{(0)}$.
2. The Left side of the criterion is free energy. Which is log of something. Write the expression for something. Bring in the Negative Log likelihood in that. Use the taylor approximation. Combine the expressions.

§4 Free parameters

Understand what is meant by "minimally singular case". What is meant by " after some change of basis"? Take an example and write the steps of this change of basis. *Hint:* Think of eigen values and vectors again.

§5 Get the feel

Cook some examples of Singularities which are free parameters and which are not. Atleast two each. Also obviously show the math of the Hessian/ FIM

Get the K v/s w plots for each of these. I want to see interesting shapes :)

§6 Learning Coefficient

Get the plot of $V(\varepsilon)$ for the examples above (for different epsilons set).
What does this idea of taking volume benefit from compared to the earlier method. Try to be mathematically rigorous.

When can the dimensionality be "read-off" from RLCT and when not?

§7 Resolving Singularities- Algebraic geometry

1. What is normal crossing form?
2. Do singular models exhibit this?
3. If we could convert the singularities to some other space such that they are analytically well defined over there (the derivaties are well-defined) using some projective geometry, such that they become a normal-crossing, won't that be great? Then we can use Watanabe's global RLCT.
4. Read up Hironaka's Resolution of Singulatities. Present it.
5. Last time you all asked what is this deal with Free energy? After this task, what do you think is the deal with it? What does it mean?